Time Series TA Session

Monday April 19, 2021

AR(1)

$$Y_t = \phi_0 + \phi_1 Y_{t-1} + a_t \qquad \{a_t\} \text{ white noise}$$

Common assumption $\{a_t\}$ iid, ind.

instead we want $\{a_t\}$ to "support" Volatility Clustering

$$\boxed{\begin{array}{l} \text{ARCH(1) model} \quad a_t = \sigma_t \varepsilon_t \qquad \{\varepsilon_t\} \text{ iid } (N(0,1)) \\ \qquad\qquad\qquad\qquad\qquad\qquad E(\varepsilon_t) = 0 \quad Var(\varepsilon_t) = 1 \\ \sigma_t = \sqrt{\omega + \alpha a_{t-1}^2} \qquad \sigma_t \text{ conditional Volatility} \end{array}}$$

Y_t = mean model + volatility model

AR(1) ARCH(1)

ARCH (1) is a white noise process

$$E(Y_t) = \frac{\phi_0}{1-\phi_1}, \quad Var(Y_t) = \frac{Var(a_t)}{1-\phi_1^2}$$

Autocorrelation $\rho_Y(h) = \phi_1^{|h|}$

Now consider $\{a_t\}$, $\mathcal{F}_t = \sigma\{a_s, s \leq t\}$

$a_t = \varepsilon_t \sigma_t$, $\sigma_t = \sqrt{\omega + \alpha a_{t-1}^2}$ $\quad \sigma_t$ is $\mathcal{F}_{t-1}$ measurable

$$E[a_t] = E[\varepsilon_t \sigma_t] = E(E(\varepsilon_t \sigma_t | \mathcal{F}_{t-1})) = E[\sigma_t \underbrace{E(\varepsilon_t | \mathcal{F}_{t-1})}_{=0}] = 0$$

$$Var(a_t) = E(a_t^2) = E(\sigma_t^2 \varepsilon_t^2) = E(\varepsilon_t^2 (\omega + \alpha a_{t-1}^2))$$

$$= \dots = \underbrace{E(\varepsilon_t^2)}_{=1} (\omega + \alpha E(a_{t-1}^2)) = \omega + \alpha E(a_{t-1}^2)$$

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$$E(a_t^2) = \omega + \alpha E(a_{t-1}^2)$$

$$\Rightarrow E(a_t^2) = \frac{\omega}{1-\alpha} \qquad Var(a_t) = \frac{\omega}{1-\alpha}$$

Important results

(1) $\{a_t\}$ is a white noise process

(2) $\{a_t^2\}$ is an AR(1) process

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t$$

$\{\nu_t\}$ is a white noise process

$(a_t = \sigma_t \varepsilon_t)$

$$\begin{aligned} a_t^2 &= \sigma_t^2 + a_t^2 - \sigma_t^2 \\ &= \omega + \alpha a_{t-1}^2 + \sigma_t^2 \varepsilon_t^2 - \sigma_t^2 \\ &= \omega + \alpha a_{t-1}^2 + \underbrace{\sigma_t^2(\varepsilon_t^2 - 1)}_{\nu_t} \end{aligned}$$

ν_t is white noise

$\nu_t = \sigma_t^2(\varepsilon_t^2 - 1)$

$E\nu_t^2 = E\left(\sigma_t^4\left(\varepsilon_t^4 - 2\varepsilon_t^2 + 1\right)\right)$ =

we need a further assumption $E(\varepsilon_t^4) < \infty$

$\varepsilon_t \sim N(0,1)$ $E(\varepsilon_t^4) = 3$

Because $\{a_t^2\}$ is an AR(1) process

$$a_t^2 = \omega + \alpha a_{t-1}^2 + \nu_t \quad (\alpha > 0)$$

we can look at $\{a_t^2\}$ and determine if there are ARCH effects (ie have heteroscedasticity)

$\Longleftrightarrow \alpha > 0$

Test for ARCH effects, Consider $\{a_t^2\}$

Look at its ACF, is it 0 or non-zero

Engel test for ARCH effects, Ljung Box

Even in the more general GARCH model

$\{a_t\}$ white noise - can use it with a standard mean model like AR, MA, ARMA, ARIMA

$\{a_t^2\}$ is an ARMA models

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in Lecture 9 - Arch/Garch and Stylized facts doesn't solve leverage effect because $\sigma_t^2 = \omega + \alpha a_t^2$

There are many variations to GARCH to "patch it up"

Focus $\{a_t^2\}$